Tech stack

Provenant's core runs on **the Python standard library alone: zero runtime dependencies**, nothing to pip-install beyond the wheel, and no model in the decision path.

AT A GLANCE

Python 3.10+

35 modules · ~7,900 lines
0 runtime deps · 235 tests

00 Layers

Four layers, one rule: **every path to a real tool goes through the same deterministic check**, and the security core never imports anything outside the standard library.

L1 · ENTRY	L2 · DECIDE	L3 · STATE	L4 · SURFACE
Agents	**Policy core**	**Proof**	**People**
MCP clients via stdio proxy, Claude Code via hooks, our gated MCP tool server	Guard + PolicyEngine in plain Python. ALLOW / DENY / REQUIRE_APPROVAL, fail closed	SQLite grants and approvals, HMAC hash-chained audit log, pinned manifests	CLI, local ops console, product site, phone alert for risky email

01 Core engine

Python `3.10-3.13`	Language for the whole security core and CLI. Packaged with setuptools; `dependencies = []`
re · unicodedata	Shell, path, secret and PII pattern rules; Unicode normalisation
codecs · base64 · binascii	Canonicalise obfuscated text: base64, hex, zero-width, tag chars, homoglyphs
hashlib · hmac · secrets	Single-use call-bound grants, audit chain, manifest pins, approval codes
sqlite3	Authorization state: WAL mode, BEGIN IMMEDIATE, conditional updates
socket · ssl · ipaddress	Netguard: decimal/hex/octal/IPv6-mapped IPs, resolve-then-pin, redirect checks
subprocess · shlex	Launch the real MCP server behind the proxy; tokenise shell commands
http.server · urllib	Local HTTP API + dashboard host on 127.0.0.1:8765
argparse · threading	`provenant` CLI; cross-process lock on the shared audit chain
ctypes (Windows)	DPAPI encryption for mail/phone settings; 0600 files elsewhere

02 Integrations

Model Context Protocol	JSON-RPC 2.0 over stdio. Proxy gates `tools/call`; HMAC-pinned tool definitions refuse rug pulls
Claude Code hooks	PreToolUse, PostToolUse, UserPromptSubmit. DENY blocks, approval becomes Claude's own ask
MCP clients	Claude Desktop, Cursor, VS Code: `provenant setup` wraps their configs with backup + rollback
Gated tool server	`provenant serve`: read_file, list_directory, search_text, run_tests, write_file, http_get
Bolna voice API	Outbound alert call for a high-risk email; placed only with `--call` and a yes
Anthropic Claude API	Optional live agent mode (`claude-opus-5-5`). Tested only with a fake client

WHY No deps

A firewall's own supply chain is attack surface. Keeping the core on the standard library means **nothing third-party sits in the decision path**, installs are offline-friendly, and every line that decides ALLOW or DENY is ours to audit.

03
Frontend

Product site, onboarding wizard and ops console. **Built once with esbuild and committed**, so the server needs no Node at runtime.

React `18.3.1`	UI for site, console and `/getting-started`
Motion `11.11.17`	Section reveals, layout transitions, DENY stamps
anime.js `3.2.2`	Page loader, two-lane attack replay, scroll marquee
esbuild `0.23.1`	Bundles `web/src` to `dashboard/app.js`
Fonts (OFL, vendored)	Bricolage Grotesque, Geist, Geist Mono. Served locally, no CDN
Plain HTML/CSS/JS	Ops console (`ops.js`, `ops.css`): zero framework

04
Delivery

Vercel	Hosted demo: static `dashboard/` + the stdlib server as a `@vercel/python` function
Wheel + sdist	setuptools build; installs with pip or pipx; dashboard ships inside the wheel
One-line installers	`install.sh` and `install.ps1`, wheel pinned by SHA-256
Supply chain	SHA256SUMS, CycloneDX SBOM, build provenance attestation
GitHub Actions	CI on Ubuntu, Windows, macOS × Python 3.10 / 3.13; actions pinned to SHAs NOT YET RUN
Windows launchers	`run.bat`, `test.bat`, `find_python.bat` (Store + python.org installs)

05
Security primitives

PRIMITIVE	BUILT WITH	GUARANTEES
Execution grant	`HMAC-SHA256 + SQLite`	runs once, only for that exact call
Audit log	`SHA-256 chain + HMAC`	any edit detected; head anchored externally
Tool manifests	`HMAC pin on first use`	changed definitions withheld
Approvals	`TTL + fingerprint + TTY`	human-only, expire, single use
Browser surface	`self-only CSP, Host check`	no third-party scripts or origins

06
Testing & evaluation

unittest	235 tests incl. a real local MCP server, clean-venv wheel install, packaging, CLI
evals/run_eval.py	225 attacks across 18 categories, 53 benign actions, 32 scenarios, 9 live proxy checks
Self red-team	Round 2: 100 new attacks written to break v8; 41 gaps found and fixed
evals/token_bench.py	Tool-result tokens 52,432 → 14,541 (−72%), one scripted workload
Playwright + Chromium	Browser walkthroughs at 1440 / 820 / 390 px: zero console errors
Fuzzer	90 mutation variants (6 seeds × 15 classes) plus false-positive corpus

07
Optional model

An advisory classifier exists but is **off by default, escalate-only, and not trained**. It can make a decision stricter; it can never allow anything.

Training workflow	PyTorch, Transformers, PEFT (LoRA), TRL, datasets, bitsandbytes, accelerate
Status	Scripts and config in `model/`; no model trained or hosted PLANNED

08
Launch kit

Deck	HTML slides rendered by Playwright, packaged with pptxgenjs
Video	three.js scenes rendered with Bun, ffmpeg mux, ElevenLabs voiceover